

AERIS CHECKPOINT — AERO SYSTEM COMPLETE, MOVING TO MASS/CG & DYNAMICS FOUNDATION

CONTEXT — WHAT AERIS IS

AERIS is a modular aircraft design software platform, not a collection of scripts.

It is structured as:

- framework layer (interfaces, models)
- implementation layer (geometry generators, solvers)
- pipeline/orchestration layer (runs, sweeps)
- CLI layer (user interaction)

The system behaves as a deterministic engineering pipeline:

geometry → solver → structured outputs → datasets → future ML and certification workflows.

GOAL:

Build a control-aware, stability-aware, dataset-ready aircraft design system for BWB UAVs.

CURRENT STATE — AERO SYSTEM (COMPLETED)

The aerodynamic subsystem is fully functional and structured.

Capabilities:

1. Geometry → AVL pipeline
 - Deterministic geometry generation (bwb_segmented_v1)
 - Export to AVL format
 - Geometry artifacts saved per run
2. Flight condition definition
 - α , β
 - p , q , r
 - velocity, altitude
 - Defines linearization point
3. Solver execution

- AVL via AeroSandbox
- Panel method
- Full raw outputs stored (no information loss)

4. Parsing into structured model

Outputs include:

- CL, CD, Cm, CY, Cl, Cn
- Stability-axis derivatives (CLa, Cma, Cnb, etc.)
- Rate derivatives (CLq, Cm_q, Clp, etc.)
- Body-axis derivatives (CXu, CZu, etc.)

5. Derived metrics

- L/D
- span efficiency
- neutral point (Xnp)
- spiral metric

6. Output model

- AeroResult (clean structured model)
- aero_result.json per run
- Includes metadata, artifacts, solver configuration

7. CLI capabilities

Single run:

- aeris aero run
- aeris aero inspect

Sweep:

- aeris aero sweep
- aeris aero sweep-inspect
- aeris aero sweep-case-inspect

8. Sweep system

- Multi-condition evaluation

- Structured manifest
- Case-level inspection
- Fully reproducible

IMPORTANT:

AERIS aero computes linear aerodynamic derivatives around operating points.
It is NOT a flight simulator.

WHAT IS EXPLICITLY NOT INCLUDED YET

- mass properties (mass, CG, inertia)
- trim formulation
- state-space models
- eigenvalue / mode analysis
- control surface modeling

These are intentionally deferred.

NEXT PHASE — MASS / CG / TRIM / DYNAMICS FOUNDATION

We now extend AERIS from:

“aerodynamic evaluator”

→ to

“aircraft dynamics-aware system”

This phase must include:

1. Mass properties model
 - mass
 - center of gravity (CG)
 - inertia placeholders (schema-ready)
2. Integration with aero outputs
 - attach CG to neutral point (Xnp)
 - compute static margin
 - prepare for stability interpretation
3. Operating condition + trim concept
 - define what is fixed vs solved

- prepare interface for trim (no full solver yet)
4. Derived stability metrics
 - static margin
 - CG-relative metrics
 - interpretation-ready outputs
 5. State-space preparation layer
 - connect derivatives + mass properties
 - define structure for future eigenvalue analysis
 6. Clean API design
 - do NOT hack into existing aero models
 - extend via new models/modules
 - maintain separation of concerns

CONSTRAINTS

- Keep architecture modular and extensible
- No shortcuts or script-style logic
- Everything must be dataset-ready and ML-compatible
- Preserve traceability (important for certification mindset)

YOUR TASK

Design and implement the MASS / CG / DYNAMICS FOUNDATION:

- Define new models (mass, CG, inertia placeholders)
- Define how they connect to AeroResult
- Add derived stability metrics (e.g., static margin)
- Extend CLI if needed (cleanly)
- Keep compatibility with existing aero pipeline

Do NOT implement:

- full trim solver
- control surface modeling
- eigenvalue analysis

Only build the foundation correctly.

IMPORTANT MINDSET

We are not adding features.

We are building a coherent aircraft design system.

Every decision must support:

- future MDAO
- surrogate modeling
- stability-aware datasets
- certification-oriented workflows

If you want, next step I can pre-design the exact data models (MassProperties, OperatingPoint, StabilityMetrics) so you don't waste time guessing.

```
(.venv) (base) mike@mike:~/Desktop/Start_Up/Code/v.0.1_Project$ tree -L 4
```

```
.
```

```
├── configs
```

```
|   ├── geometry
```

```
|   |   ├── baseline_bwb_25.yaml
```

```
|   |   ├── baseline_bwb.yaml
```

```
|   |   └── wing_bwb.yaml
```

```
|   └── smoke
```

```
|       ├── bad.yaml
```

```
|       └── dev.yaml
```

|— data

| |— datasets

| | |— bwb25_demo5

| | |— configs

| | |— dataset_manifest.json

| | |— failures.csv

| | |— geometry

| | |— logs

| | |— metadata.csv

| |— debug

| | |— test_aero_sweep

| | | |— aero_sweep_manifest.json

| | | |—
case_0000_a+0.000_b+0.000_V28.000_alt1500.0_p+0.0000_q+0.0000_r+0.0000

| | | |—
case_0001_a+0.000_b+0.000_V28.000_alt1500.0_p+0.0000_q+0.0500_r+0.0000

| | | |—
case_0002_a+2.000_b+0.000_V28.000_alt1500.0_p+0.0000_q+0.0000_r+0.0000

| | | | —
case_0003_a+2.000_b+0.000_V28.000_alt1500.0_p+0.0000_q+0.0500_r+0.0000

| | | | —
case_0004_a+4.000_b+0.000_V28.000_alt1500.0_p+0.0000_q+0.0000_r+0.0000

| | | | —
case_0005_a+4.000_b+0.000_V28.000_alt1500.0_p+0.0000_q+0.0500_r+0.0000

| | | — test_aero_sweep_manifest.json

| | | — test_sweep_geometry

| | | — airfoils

| | | — control_points.csv

| | | — geometry_summary.json

| | | — planform_sections.csv

| | | — section_3d.csv

| | — interim

| | — processed

| | — raw

| | — runs

| | — 20260321_191326_aero_baseline_bwb_25

| | | — aero

- | | | — aero_manifest.json
- | | | — baseline_bwb_25.yaml
- | | | — geometry
- | | — 20260321_191342_aero_baseline_bwb_25
- | | | — aero
- | | | — geometry
- | | — 20260321_191411_aero_geometry
- | | | — aero
- | | | — geometry
- | | — 20260321_193252_aero_baseline_bwb_25
- | | | — aero
- | | | — geometry
- | | — 20260321_193500_aero_baseline_bwb_25
- | | | — aero
- | | | — aero_manifest.json
- | | | — baseline_bwb_25.yaml
- | | | — geometry

| └─ 20260321_193519_aero_baseline_bwb_25

| | └─ aero

| | └─ aero_manifest.json

| | └─ baseline_bwb_25.yaml

| | └─ geometry

| └─ 20260321_193529_aero_20260321_185956_aero_baseline_bwb_25

| | └─ aero

| | └─ geometry

| └─ 20260321_193550_aero_geometry

| | └─ aero

| | └─ geometry

| └─ 20260321_193600_aero_geom_00001

| | └─ aero

| | └─ geometry

| └─ 20260321_193742_aero_geometry

| | └─ aero

| | └─ aero_manifest.json

- | | └─ geometry
- | └─ 20260321_193801_aero_20260321_193519_aero_baseline_bwb_25
- | | └─ aero
- | | └─ aero_manifest.json
- | | └─ geometry
- | └─ 20260321_194238_aero_geom_00001
- | | └─ aero
- | | └─ aero_manifest.json
- | | └─ geometry
- | └─ 20260321_194249_aero_geom_00005
- | | └─ aero
- | | └─ aero_manifest.json
- | | └─ geometry
- | └─ 20260321_195721_aero_baseline_bwb_25
- | | └─ aero
- | | └─ aero_manifest.json
- | | └─ baseline_bwb_25.yaml

- | | └─ geometry
- | └─ 20260321_195752_aero_baseline_bwb_25
- | | └─ aero
- | | └─ aero_manifest.json
- | | └─ baseline_bwb_25.yaml
- | | └─ geometry
- | └─ 20260321_200756_aero_baseline_bwb_25
- | | └─ aero
- | | └─ aero_manifest.json
- | | └─ baseline_bwb_25.yaml
- | | └─ geometry
- | └─ 20260321_201523_aero_baseline_bwb_25
- | | └─ aero
- | | └─ aero_manifest.json
- | | └─ baseline_bwb_25.yaml
- | | └─ geometry
- | └─ 20260321_204617_aero_baseline_bwb_25

| | └─ aero

| | └─ aero_manifest.json

| | └─ baseline_bwb_25.yaml

| | └─ geometry

| └─ 20260321_205238_aero_baseline_bwb_25

| | └─ aero

| | └─ aero_manifest.json

| | └─ baseline_bwb_25.yaml

| | └─ geometry

| └─ 20260322_123541_aero_baseline_bwb_25

| | └─ aero

| | └─ aero_manifest.json

| | └─ baseline_bwb_25.yaml

| | └─ geometry

| └─ 20260322_130835_aero_baseline_bwb_25

| | └─ aero

| | └─ aero_manifest.json

- | | | — baseline_bwb_25.yaml
- | | | — geometry
- | | — 20260322_130957_aero_baseline_bwb_25
- | | | — aero
- | | | — aero_manifest.json
- | | | — baseline_bwb_25.yaml
- | | | — geometry
- | | — 20260322_131412_aero_baseline_bwb_25
- | | | — aero
- | | | — aero_manifest.json
- | | | — baseline_bwb_25.yaml
- | | | — geometry
- | | — 20260322_131850_aero_baseline_bwb_25
- | | | — aero
- | | | — aero_manifest.json
- | | | — baseline_bwb_25.yaml
- | | | — geometry

| └─ 20260322_133516_aero_baseline_bwb_25

| | └─ aero

| | └─ aero_manifest.json

| | └─ baseline_bwb_25.yaml

| | └─ geometry

| └─ 20260322_133625_aero_baseline_bwb_25

| | └─ aero

| | └─ aero_manifest.json

| | └─ baseline_bwb_25.yaml

| | └─ geometry

| └─ 20260322_133722_aero_baseline_bwb_25

| | └─ aero

| | └─ aero_manifest.json

| | └─ baseline_bwb_25.yaml

| | └─ geometry

| └─ 20260322_134449_aero_baseline_bwb_25

| | └─ aero

- | | | — aero_manifest.json
- | | | — baseline_bwb_25.yaml
- | | | — geometry
- | | — 20260322_135057_aero_baseline_bwb_25
- | | | — aero
- | | | — aero_manifest.json
- | | | — baseline_bwb_25.yaml
- | | | — geometry
- | | — 20260322_141425_aero_baseline_bwb_25
- | | | — aero
- | | | — aero_manifest.json
- | | | — baseline_bwb_25.yaml
- | | | — geometry
- | | — 20260322_143250_aero_sweep_baseline_bwb_25
- | | | — aero_sweep
- | | | — aero_sweep_manifest.json
- | | | — baseline_bwb_25.yaml

- | | └─ geometry
- | └─ 20260322_161739_aero_baseline_bwb_25
- | | └─ aero
- | | └─ aero_manifest.json
- | | └─ baseline_bwb_25.yaml
- | | └─ geometry
- | └─ dev
- | └─ archive
- | └─ planform_generator.py
- | └─ documents
- | | └─ Aeris Architecture Guide.pdf
- | | └─ aeris_cheat_sheet.pdf
- | | └─ aeris_checkpoint_5_cli_commands.pdf
- | | └─ aeris_checkpoint_5_explanation.docx
- | | └─ aeris_checkpoint_5_explanation.pdf
- | | └─ aeris_checkpoint_6_explanation.pdf
- | | └─ aeris_checkpoint_1_report.pdf

- | |— aeris_checkpoint_2_report.pdf
- | |— aeris_checkpoint_3_report.docx
- | |— aeris_checkpoint_3_report.pdf
- | |— aeris_checkpoint_4_report.pdf
- | |— aeris_checkpoint_5_report.pdf
- | |— aeris_checkpoint_6.5_report.pdf
- | |— aeris_checkpoint_6_report.pdf
- | |— aeris_checkpoint_7_report.pdf
- | |— aeris_updated_cheat_sheet.docx
- | |— aeris_updated_cheat_sheet.pdf
- | |— Calendar.docx
- | |— Guidelines.docx
- | |— Instructions_for_newbies.docx
- | |— Master_Plan.docx
- | |— README.md
- | |— SW_Plan (Copy).docx
- | |— SW_Plan.docx

- |— geom_test.py
- |— __pycache__
- | └─ geom_test.cpython-313.pyc
- |— pyproject.toml
- |— README.md
- |— ROADMAP.md
- |— scripts
- | |— bwb_segmented_v1
- | | |— design_space_inspector.py
- | | |— __pycache__
- | | | |— design_space_inspector.cpython-313.pyc
- | | | |— run_single_case_and_plot_3d.cpython-313.pyc
- | | | |— test_lhs.cpython-313.pyc
- | | | |— test_lhs_to_geometry.cpython-313.pyc
- | | | └─ visualize_airplane.cpython-313.pyc
- | | |— run_single_case_and_plot_3d.py
- | | |— test_lhs.py

- | | └─ test_lhs_to_geometry.py
- | | └─ visualize_airplane.py
- | └─ generic
- | | └─ audit_dataset.py
- | | └─ check_architecture_integrity.py
- | | └─ check_bad_config_failure.py
- | | └─ check_large_dataset_sanity.py
- | | └─ check_metadata_regeneration.py
- | | └─ check_seed_variation.py
- | | └─ __pycache__
- | | └─ audit_dataset.cpython-313.pyc
- | | └─ check_architecture_integrity.cpython-313.pyc
- | | └─ check_bad_config_failure.cpython-313.pyc
- | | └─ check_large_dataset_sanity.cpython-313.pyc
- | | └─ check_metadata_regeneration.cpython-313.pyc
- | | └─ check_seed_variation.cpython-313.pyc
- | └─ section_study_pick_default.py

- | |─ section_study.py
- | |─ utilities
- | |─ make_dataset_gif.py
- | |─ __pycache__
- | |─ make_dataset_gif.cpython-313.pyc
- |─ src
- | |─ aeris
- | | |─ aero
- | | |─ base.py
- | | |─ __init__.py
- | | |─ io.py
- | | |─ models.py
- | | |─ __pycache__
- | | |─ registry.py
- | | |─ solvers
- | | |─ sweep_io.py
- | | |─ sweep.py

| | | └─ sweep_run.py

| | | └─ validation.py

| | | └─ views.py

| | └─ cli.py

| | └─ commands

| | | └─ aero.py

| | | └─ dataset.py

| | | └─ geometry.py

| | | └─ __init__.py

| | | └─ pipeline.py

| | | └─ __pycache__

| | | └─ version.py

| | └─ common

| | | └─ config.py

| | | └─ __init__.py

| | | └─ logging_utils.py

| | | └─ paths.py

| | | └─ __pycache__

| | └─ dataset

| | | └─ dataset_run.py

| | | └─ __init__.py

| | | └─ inspect.py

| | | └─ io.py

| | | └─ metadata.py

| | | └─ models.py

| | | └─ __pycache__

| | | └─ sampling

| | └─ generators

| | | └─ bwb_segmented_v1

| | | └─ __init__.py

| | | └─ __pycache__

| | └─ geometry

| | | └─ base.py

| | | └─ config_resolver.py

| | | |— generators

| | | |— __init__.py

| | | |— __pycache__

| | | |— registry.py

| | | |— __init__.py

| | | |— pipeline

| | | |— geometry_run.py

| | | |— __init__.py

| | | |— __pycache__

| | | |— smoke.py

| | | |— __pycache__

| | | |— cli.cpython-313.pyc

| | | |— __init__.cpython-313.pyc

| | | |— aeris.egg-info

| | | |— dependency_links.txt

| | | |— entry_points.txt

| | | |— PKG-INFO

```
|   |— requires.txt  
  
|   |— SOURCES.txt  
  
|   └— top_level.txt  
  
└— tests
```

154 directories, 162 files

(.venv) (base) mike@mike:~/Desktop/Start_Up/Code/v.0.1_Project\$